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PROFESSIONAL SUMMARY Innovative Data Scientist and AI Engineer with 8+ years of experience specializing in scalable infrastructure and machine learning. Proven track record of deploying production-grade AI agents and optimizing large-scale data pipelines. Expert in leveraging LLMs, vector databases, and image processing to drive multi-million dollar cost savings and operational efficiency. Proficient in modern AI development workflows including Claude Code development, Claude Code MCPs, and Cursor IDE for rapid engineering. Experienced in fine-tuning Hugging Face open-source models, utilizing Ollama for local model optimization, OpenClaw for agentic frameworks, and Databricks MLflow for managing the machine learning lifecycle and Snowflake's Cortex.

TECHNICAL SKILLS

- **AI & Machine Learning:** LLM Configuration & Fine-Tuning, AI Agent Orchestration (LangChain/AutoGPT), Image Processing (Yolov3, OpenCV, PyTorch, TensorFlow), Natural Language Processing (NLP), Text Embeddings & Sentence Chunking (Vector Databases), Reinforcement Learning.
- **Programming Languages & Frameworks:** Python (Advanced), PySpark, NumPy, Pandas, Seaborn, ML Libraries, TensorFlow, PyTorch, Sklearn (Scikit Learn), JavaScript/TypeScript, React.js, SQL (Advanced).
- **Cloud & Infrastructure:** AWS (EC2, EKS, ECR, S3, Athena, QuickSight, IAM, Glue, Kinesis, Redshift, SQS), Azure (Container Registry), Kubernetes, Docker, Databricks.
- **Data Engineering & Analytics:** ETL Pipeline Development, Machine Learning Architecture, Anomaly Detection Systems, Real-time Data Processing, Data Warehouse Architecture (Delta Live Tables, Snowflake), Scalable Large Data Pipelines, Postgres, Apache Spark, Polars, Data Lake Architecture, SFTP/Data Integration.
- **DevOps & Automation:** Continuous Deployment, Cloud Architecture, Terraform, UiPath, Robotic Process Automation (RPA), CI/CD Pipelines.
- **Visualization & Reporting:** Tableau, AWS QuickSight, Data Storytelling, Business Intelligence.

EXPERIENCE

SciPlay | Core AI/Data Engineer *January 2023 - Present*

- **AI Agent Production:** Architected and deployed autonomous AI agents into the production environment to automate complex decision-making workflows, enhancing operational responsiveness.
- **Infrastructure Optimization:** Generated over $7,000,000 in storage cost savings through the implementation of advanced data lifecycle management and compression algorithms.
- **LLM Implementation:** Managed the configuration and fine-tuning of Large Language Models (LLMs) to improve model accuracy for domain-specific tasks.
- **Search & Retrieval:** Developed high-performance retrieval systems utilizing Vector Databases to enable semantic search capabilities across massive datasets.

Apple | Machine Learning Engineer - AI Ops (Contract) *April 2022 - September 2024*

- Scaled anomaly detection system from 10 to 200 pipelines while reducing SLA times from 20 to 1 minute.
- Developed a self-serving API enabling automated pipeline onboarding for customized anomaly detection.
- Implemented optimized cache storage and ensemble machine learning models to enhance detection speed.
- Built a data classification system for unsupervised machine learning models, preventing DDOS attacks on Apple TV and fraudulent Apple Pay transactions (iMessage, FindMe, Mail, AppStore).
- Utilized AI-driven image processing techniques (OpenCV) to automate visual data verification, reducing manual oversight by 40%.

Saatva | Data Engineer *June 2021 - July 2022*

- Engineered ETL pipelines integrating multiple advertising platforms (Facebook, Twitter, Snapchat) into AWS Athena.
- Developed automated data ingestion solutions using Python and PySpark for marketing analytics.
- Implemented hourly data refresh cycles using AWS Glue scripts for real-time marketing insights.

AT&T | Full Stack Engineer (Contract) *January 2021 - April 2022*

- Led development of Northstar, a full-stack application consolidating multiple legacy data sources into a single source of truth.
- Architected a data validation system to identify mismatched types and synchronization issues.
- Designed and implemented ETL processes based on user-defined forms and requirements.

Degree Analytics | Data Scientist *October 2019 - December 2020*

- Analyzed student success patterns using IoT sensor data and demographic information to identify key success factors.
- Managed AWS infrastructure, including S3, SFTP, and IAM controls for university data onboarding.
- Created data visualizations using AWS QuickSight to communicate insights to university stakeholders.

Merck | Associate Technologist *January 2019 - October 2019*

- Developed MVP projects, including an innovative LLM-powered (Rasa) chatbot as an application for SharePoint integration.
- Led international RPA initiatives to automate business processes.
- Collaborated with cross-functional teams to deliver rapid prototypes for innovation projects.

Merck | Robotic Process Automation (Contract) *October 2018 - January 2019*

- Automated SAP discount verification process for 1000 vaccination purchases, resulting in $250,000+ savings within the first 6 weeks.

- Implemented OCR machine learning automation solutions for international teams.

Degree Analytics | Data Scientist (Intern) *May 2018 - August 2018*

- Created a facility usage heat map feature using PySpark and Seaborn for visualizing campus activity.
- Conducted machine learning research to address university-specific feature requests.

EDUCATION

- **Master of Science, Data Science** | Eastern University | 2025
- **Bachelor of Science, Mathematics** | Huston-Tillotson University, Austin, Texas | 2019
- **Data Engineering/Data Scientist Certificate** | The University of Texas, Austin
- **Associate of Science - Mathematics** | Austin Community College
- **Databricks Certified Data Engineer Associate** | Credential ID 70165650

PROFESSIONAL PROJECTS

- **Automated Data Glaciering System | SciPlay:** Engineered an intelligent data archiving system that automatically moves aging data to AWS Glacier. Reduced storage costs by $20,000/month through optimized data lifecycle management. The project was adopted by Databricks as a platform feature.
- **Enterprise Anomaly Detection Platform | Apple:** Architected a scalable system processing 300+ pipelines with a 1-minute SLA. Reduced false positives by 40% through advanced data classification (fine-tuning).
- **Northstar Data Integration Platform | AT&T:** Built a full-stack application consolidating multiple enterprise databases. Significantly improved data accuracy across enterprise systems ($100K/monthly savings).
- **LLM-Powered SharePoint Chatbot | Merck:** Developed a customizable chatbot for SharePoint integration. Adopted company-wide as a standard solution for department documentation.
- **Campus Activity Heat Map System | Degree Analytics:** Developed a real-time visualization system for campus facility usage, processing IoT sensor data to track student movement patterns.

PERSONAL PROJECTS

- **Custom RAG-Enhanced Language Model (PocketChat.io):** Developed a system enabling users to create personalized LLMs using their own PDF documentation. Implemented RAG (Retrieval-Augmented Generation) architecture for processing 1,000+ page documents and built a vector database integration for semantic search and context retrieval.
- **Deep Learning, Reinforcement Learning on Market Data:** Built a software tool to automatically trade based on Q-values from reinforcement learning models. Built WebSockets to handle incoming anomalies and backtesting on large data for sentient AI to understand market and streaming data decision-making.
- **Stock Market Straddle Timing Predictor:** Engineered a neural network achieving 98% success rate in straddle timing predictions. Implemented volume spike detection and anomaly recognition for options contracts.